Quantum Information and Computation
Course ID: 021662 MATH F266 STUDY PROJECT

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Academic Year 2025-2026

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1 Single Systems

1.1 Thursday, Jan 29: First lecture

1) [**Classical States**]

Take a physical system termed X.
∃ certain values that X can hold depending on the system under consideration. Assume that X has a finite number of possible values. These values are known as classical states.

Example 1. $X = \{0, 1, 2, 3, 4, 5\} \implies$ *X can hold states 1,2,3,4,5.*

2)**Probabilistic State**

Definition 1 (Probabilistic State)**.** *A probabilistic state describes the probability distribution over the classical states of X. It is represented by a column vector whose entries correspond to the associated probabilities. Also, the sum of values of the column vector is 1 because* $\sum Pr = 1$.

Example 2. *For X in Example 1, the vector can be taken as*

$$\mathbf{p} = \begin{pmatrix} 1/10 \\ 2/10 \\ 2/10 \\ 2/10 \\ 2/10 \\ 1/10 \end{pmatrix} \qquad \begin{array}{l} Pr(0) = 1/10 \\ Pr(1) = 2/10 \\ Pr(2) = 2/10 \\ Pr(3) = 2/10 \\ Pr(4) = 2/10 \\ Pr(5) = 1/10 \end{array}$$

3)**Dirac Notation**

The Dirac (bra–ket) notation is a compact mathematical language used in quantum mechanics to describe quantum states. By assigning an order to the classical state set, quantum states can be written as column vectors, called kets.

Example 3. *In $X = \{0, 1\}$ we get,*

Table 1: Ordering of classical states

Classical state	**Order**
0	0
1	1

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \begin{array}{l} \textit{1 because 0 is the element of order } 0. \\ \textit{0 because 0 is not an element of order } 1. \end{array}$$

Similarly,

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A bra is the dual or conjugate transpose of the ket.

$$\langle 0 \,| = \begin{pmatrix} 1 & 0 \end{pmatrix} \quad \langle 1 \,| = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

4)**Relating $|a\rangle$ to the Probabilistic State**

Suppose in $X = \{0, 1\}$ we have Pr(0)=1/3 and Pr(1)=2/3 then we can write

$$\begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \end{pmatrix} = \frac{1}{3} |0\rangle + \frac{2}{3} |1\rangle$$

Remark 1.

- $$\langle a|b\rangle = \left\{ \begin{array}{ll} 1, & \textit{if } a = b, \\ 0, & \textit{if } a \neq b. \end{array} \right\}$$
- *$|a\rangle\langle b|$ is an $n \times n$ matrix whose (i, j)th entry is 1 and all other entries are 0, where a corresponds to the ith element of the classical set and b corresponds to the jth element of the classical set.*

5)**Classical Operations**

a)Deterministic operations

Every function $f : \Sigma \to \Sigma$ (classical states set to itself) describes a deterministic operation that transforms the classical state a into $f(a)$, for each $a \in \Sigma$. Given any function $f : \Sigma \to \Sigma$, there is a (unique) matrix M satisfying

$$M|a\rangle = |f(a)\rangle \qquad \text{(for every } a \in \Sigma).$$

This matrix has exactly one 1 in each column, and 0 for all other entries:

$$M(b, a) = \left\{ \begin{array}{ll} 1 & b = f(a) \\ 0 & b \neq f(a) \end{array} \right\}$$

The action of this operation is described by matrix-vector multiplication:

$$v \longmapsto Mv.$$

Example 4. $X = \{0, 1\}, \qquad f(0) = 1,\ f(1) = 0$

$$M = \begin{pmatrix} M(0,0) & M(0,1) \\ M(1,0) & M(1,1) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$M(0,0) = 0 \quad \textit{since } 0 \neq f(0) = 1,$
$M(0,1) = 1 \quad \textit{since } 0 = f(1),$
$M(1,0) = 1 \quad \textit{since } 1 = f(0),$
$M(1,1) = 0 \quad \textit{since } 1 \neq f(1) = 0.$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \Rightarrow \quad |0\rangle \mapsto |1\rangle$$
$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Rightarrow \quad |1\rangle \mapsto |0\rangle$$

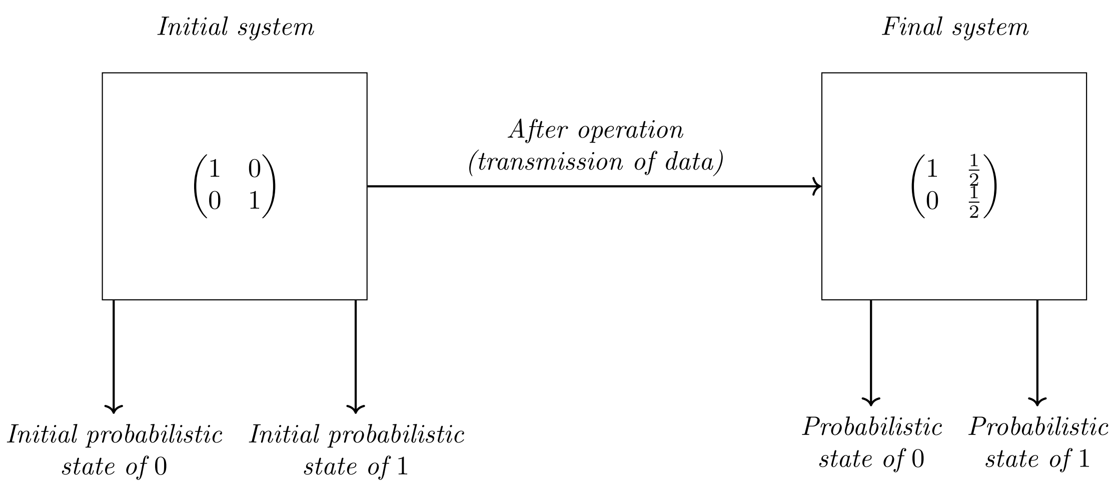
6)**Stochastic Matrix**

A stochastic matrix is a matrix whose columns represent probability distributions. In a stochastic matrix, the jth column represents the probability distribution of the output states(from the operation) when the system starts in the basis state $|j\rangle$. A matrix $A = (A_{ij})$ is called *stochastic* if

$$A_{ij} \geq 0 \quad \text{for all } i, j$$

$$\sum_i A_{ij} = 1 \quad \text{for every column } j$$

Example 5. *A stochastic matrix is used to indicate the probability distribution change during the transmission of data.*



Remark 2. *If M is a matrix (linear operator) and $|x\rangle$ is a ket, then*

$$M\big(M|x\rangle\big) = M^2|x\rangle$$

where $M^2 = M \cdot M$

1.2 Wed, Feb 12: Second Lecture

1)**Quantum States**

Suppose I is an isolated system which has a classical state set

$$\Sigma = \{1, 2, 3, 4, 5, \ldots, n\}.$$

Then the quantum state of the system can be written as

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \qquad \alpha_i \in \mathbb{C}.$$

Here $|\alpha_i|^2$ represents the probability of each classical state $|i\rangle$, and

$$\sum_{i=1}^{n} |\alpha_i|^2 = 1.$$

Example 6. $\Sigma = \{0, 1\}$

$$|\psi\rangle = \begin{pmatrix} \frac{1+2i}{3} \\ \frac{2}{3} \end{pmatrix} = \frac{1+2i}{3}\,|0\rangle + \frac{2}{3}\,|1\rangle$$

$$P(0) = \left|\frac{1+2i}{3}\right|^2 = \left(\frac{1}{3}\right)^2 + \left(\frac{2}{3}\right)^2 = \frac{5}{9}$$

$$P(1) = \left|\frac{2}{3}\right|^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$$

$$\sum_i |\alpha_i|^2 = \frac{5}{9} + \frac{4}{9} = 1$$

2)**Norm of a Vector**

Suppose there exist a vector v such that

$$v = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \qquad \alpha_i \in \mathbb{C}, \qquad \sum_{i=1}^{n} |\alpha_i|^2 = 1.$$

$$\|v\| = f : \mathbb{C}^n \to \mathbb{R}^+$$

Properties:

$$\begin{aligned} &\text{(i)} \quad \|v\| \geq 0, \qquad \|v\| = 0 \iff v = 0, \\ &\text{(ii)} \quad \|\alpha v\| = |\alpha|\,\|v\|, \qquad \alpha \in \mathbb{C}, \\ &\text{(iii)} \quad \|u + v\| \leq \|u\| + \|v\| \quad \text{(Triangle inequality)}. \end{aligned}$$

For quantum states representation, we follow the Euclidean norm:

$$\|v\| = \sqrt{\sum_{i=1}^{n} |\alpha_i|^2}.$$

3)**Relation of Inner Product and Norm of a Vector**

$$|\psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \vdots \\ \alpha_n \end{pmatrix}, \qquad \langle\psi| = (\overline{\alpha_1}, \overline{\alpha_2}, \ldots, \overline{\alpha_n}),$$

where $\alpha_i \in \mathbb{C}$, $\overline{\alpha_i}$ is the complex conjugate of α_i,

$$\Rightarrow \; \langle\psi|\psi\rangle = (\overline{\alpha_1}, \ldots, \overline{\alpha_n}) \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \sum_{i=1}^{n} \overline{\alpha_i}\alpha_i = \sum_{i=1}^{n} |\alpha_i|^2 = \|\psi\|^2.$$

Remark 3. *When the classical state is*

$$\Sigma = \{0, 1\},$$

then the quantum state of the system is called the qubit state.

4)**Quantum Superposition**

Definition 2. *Quantum systems allow linear combination of states (unlike classical states). This principle is known as the principle of superposition.*

Remark 4. *A qubit can exist in both states:*

$$|+\rangle = \frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle, \qquad |-\rangle = \frac{1}{\sqrt{2}}\,|0\rangle - \frac{1}{\sqrt{2}}\,|1\rangle.$$

5)**Unitary Matrix**

Definition 3. *A square matrix $U \in \mathbb{C}^{n\times n}$ is unitary if*

$$U^\dagger U = UU^\dagger = I,$$

where

- $U^\dagger$ *= conjugate transpose of* U*,*
- I *= identity matrix.*

So,

$$U^{-1} = U^\dagger.$$

Unitary Matrices preserve both the inner product and the norm of the vector. Consider a unitary matrix U and a quantum state vector $|v\rangle$

$$(U|v\rangle)^\dagger = \langle v|U^\dagger$$

$$\|U|v\rangle\|^2 = \langle Uv \mid Uv\rangle = \langle v \mid U^\dagger U \mid v\rangle = \langle v \mid v\rangle = \|v\|^2 \quad \{\, U^\dagger U = I \,\}$$

6)**Hermitian Matrix**

Definition 4. *A matrix is Hermitian if $A = A^\dagger$, where $A^\dagger$ is the conjugate transpose of matrix A. $a_{ij} = \overline{a_{ji}}$ for all i, j.*

Hermitian matrices are essential for defining operations on a qubit. The eigenvalues of a Hermitian matrix are always real.

7)**Unitary Qubit Operations**

Operations of unitary matrices on qubits result in the formation of new qubits. Certain examples are-

1) NOT Gate

$$\alpha_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\alpha_x|0\rangle = |1\rangle, \qquad \alpha_x|1\rangle = |0\rangle$$

2) Bit-flip + Phase Change

$$\alpha_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\alpha_y|0\rangle = i|1\rangle, \qquad \alpha_y|1\rangle = -i|0\rangle$$

3)Phase Flip

$$\alpha_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\alpha_z|0\rangle = |0\rangle, \qquad \alpha_z|1\rangle = -|1\rangle$$

4) Hadamard Matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Lemma 1. *If T is the linear transformation of a qubit, then $T = [\, Te_1 \;\; Te_2 \;\; Te_3 \;\; \cdots \;\; Te_n \,]$*

Example 7.

$$\alpha_z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle,$$

$$\alpha_z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle.$$

Remark 5.

- *All Hermitian matrices are not unitary matrices and vice versa.*
- *If a matrix is both Hermitian and unitary, then*

$$A = A^\dagger = A^{-1} \quad \Rightarrow \quad A^2 = I.$$

8)**Hadamard Conjecture**

$$\forall\, n = 4k,\ k \in \mathbb{N},\ \exists\, H_n \in \{\pm 1\}^{n \times n} \text{ such that } H_n H_n^\dagger = nI.$$

The sylvester construction (recursive) gives matrices for $n = 2^m$. When a quantum state acts on Hadamard matrix, then $|+\rangle$ and $|-\rangle$ are formed.

$$H|+\rangle = |0\rangle, \qquad H|-\rangle = |1\rangle$$

1.3 Thursday, Feb 19: Third lecture

1)**Phase Operations**

Phase operations are operations done by

$$P_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \quad \text{where}$$

$$\text{For } \theta = 0 \quad P_0 = I$$

$$\text{For } \theta = \frac{\pi}{2} \quad P_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

$$\text{For } \theta = \frac{\pi}{4} \quad P_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & \dfrac{1+i}{\sqrt{2}} \end{pmatrix} = T \text{ (denotion)}$$

$$\text{For } \theta = \pi \quad P_\pi = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z$$

2)**Composition of Unitary Matrix**

Theorem 1. *The composition of unitary $n \in \mathbb{N}$ matrices is a unitary matrix*

Proof: Case for one matrix:

$$U^\dagger U = I$$

For two matrices,

$$(UV)^\dagger (UV) = V^\dagger U^\dagger UV = V^\dagger V = I$$

Now assume that it is true for $n-1$ matrices. Let P be the composition of the $(n-1)$ matrices and D_n be the matrix n^{th}.

$$P^\dagger P = U$$

$$(PD_n)^\dagger (PD_n) = D_n^\dagger P^\dagger P D_n = D_n^\dagger D_n = I$$

Hence proved.

Example 8.

$$H\sigma_x H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$H\alpha_z H = \alpha_x$$

$$HH\alpha_x HH = H\alpha_z H = I\alpha_x I = \alpha_x$$

$$HP_{\pi/2}H = HSH = R = \begin{bmatrix} \dfrac{1+i}{2} & \dfrac{1-i}{2} \\ \dfrac{1-i}{2} & \dfrac{1+i}{2} \end{bmatrix}$$

$$\mathit{Also} \quad R^2 = P_\pi = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

2 Multiple Systems

2.1 Thursday, Feb 19: Third Lecture

1)**Classical states**
Consider two systems X and Y with the classical state set $\sum$ and Γ. Then, the elements in the classical state of XY will be derived by the Cartesian product $\sum \times \Gamma$

Example 9. $\sum = \{0,1\}$ $\Gamma = \{0,1\}$
$\sum \times \Gamma = \{(0,0),(0,1),(1,1),(1,0)\}$

2)**Probabilistic State**
Let ϕ be a probabilistic state vector of the system of XY.

$$|\phi\rangle = \begin{pmatrix} \frac{1}{3} \\ \frac{1}{3} \\ 0 \\ \frac{1}{3} \end{pmatrix} \quad \Rightarrow \quad \begin{aligned} P(0,0) &= \tfrac{1}{3} \\ P(0,1) &= \tfrac{1}{3} \\ P(1,0) &= 0 \\ P(1,1) &= \tfrac{1}{3} \end{aligned}$$

Note: We have fixed the position of each classical state.

3)**Tensor Product**

Definition 5. *Tensor product $V \otimes W$, where V and W are vector spaces.*
Let

$$|V\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}, \qquad |W\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

Then

$$|V\rangle \otimes |W\rangle = \begin{pmatrix} \alpha_1\beta_1 \\ \alpha_1\beta_2 \\ \alpha_2\beta_1 \\ \alpha_2\beta_2 \\ \alpha_3\beta_1 \\ \alpha_3\beta_2 \end{pmatrix}$$

We can also denote

$$|V\rangle = \sum_{a\in\Sigma} \alpha_a |a\rangle, \qquad |W\rangle = \sum_{b\in\Gamma} \beta_b |b\rangle$$

where Σ and Γ are the Classical State sets for V and W respectively.
Thus,

$$|V\rangle \otimes |W\rangle = \sum_{a\in\Sigma} \sum_{b\in\Gamma} \alpha_a \beta_b \, |ab\rangle$$

Note: $|ab\rangle$ is a standard basis vector with all entries 0 except at the position corresponding to (a,b), where the entry is 1

4)**Bilinearity in Tensor Product**
Let

$$|\phi_1\rangle = \sum_i \alpha_i |a_i\rangle, \qquad |\phi_2\rangle = \sum_i \beta_i |a_i\rangle$$

(both in the same basis), and let

$$|\psi\rangle = \sum_j \delta_j |c_j\rangle.$$

Then for vectors $|\phi_1\rangle, |\phi_2\rangle, |\psi\rangle$ and scalar C,

$$\big(|\phi_1\rangle + |\phi_2\rangle\big) \otimes \big(C|\psi\rangle\big) = C\big(|\phi_1\rangle \otimes |\psi\rangle\big) + C\big(|\phi_2\rangle \otimes |\psi\rangle\big).$$

Proof:

$$\text{LHS} = \left(\sum_i (\alpha_i + \beta_i)|a_i\rangle\right) \otimes \left(\sum_j \delta_j |c_j\rangle\right)$$

$$= \sum_{i,j} (\alpha_i + \beta_i)\, \delta_j \, |a_i c_j\rangle (Definition - 5)$$

$$= \sum_{i,j} \alpha_i \delta_j |a_i c_j\rangle + \sum_{i,j} \beta_i \delta_j |a_i c_j\rangle (separate \alpha_i and \beta_i)$$

$$= |\phi_1\rangle \otimes |\psi\rangle + |\phi_2\rangle \otimes |\psi\rangle. = RHS$$

Hence proved.

Note: Let

$$|\phi\rangle = \sum_i \alpha_i |a_i\rangle, \qquad |\psi\rangle = \sum_j \beta_j |b_j\rangle.$$

Then the state of the tensor product is

$$|\Pi\rangle = |\phi\rangle \otimes |\psi\rangle = \sum_{i,j} \alpha_i \beta_j |a_i b_j\rangle.$$

Now, the inner product with a basis vector gives

$$\langle a_i b_j \,|\, \Pi\rangle = \alpha_i \beta_j,$$

i.e., the amplitude corresponding to that particular pair (i, j).

2.2 Thursday, March 6: Fourth Lecture

1)**Independent Multiple System**

The tensor product of the individual systems describes a multiple system. In general, a state of an n-partite system can be written as a summation of tensor product basis states:

$$|\Psi\rangle = \sum_{i_1, i_2, \ldots, i_n} a_{i_1 i_2 \ldots i_n} \, |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_n\rangle \,.$$

Such a state represents the most general form and may describe a dependent system.

A system is said to be independent if its state can be written as a single tensor product of individual subsystem states. That is, if there exist states $|p_1\rangle, |p_2\rangle, \ldots, |p_n\rangle$ such that

$$|\Psi\rangle = |p_1\rangle \otimes |p_2\rangle \otimes \cdots \otimes |p_n\rangle$$

In this case, the coefficients will become as follows.

$$a_{i_1 i_2 \ldots i_n} = a^{(1)}_{i_1} \, a^{(2)}_{i_2} \cdots a^{(n)}_{i_n},$$

2)**Marginal Probability**

Definition 6. *The marginal distribution of X that is the probability to observe a particular classical state $a \in X$ when just X is measured is*

$$P(X = a) = \sum_{c \in Y} P_{ac}$$

We compute the marginal probabilities by summing over Y.

Example 10. *Consider classical systems X and Y with classical states given as:*

$$X = \{0, 1\}, \quad Y = \{0, 1\}$$

Now, let Π be a multiple system formed by X and Y.

$$\Pi = \{(0,0), (1,1), (1,0), (0,1)\}$$

$$|\Pi\rangle = \frac{1}{2}|00\rangle + \frac{1}{4}|01\rangle + \frac{1}{8}|10\rangle + \frac{1}{8}|11\rangle$$

where

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = 1$$

Now, we need to define $P(X = 0)$ and $P(X = 1)$.

$$|\Pi\rangle = |0\rangle \otimes \left(\frac{1}{2}|0\rangle + \frac{1}{4}|1\rangle\right) + |1\rangle \otimes \left(\frac{1}{8}|0\rangle + \frac{1}{8}|1\rangle\right)$$

$$P(X = 0) = \sum_{c \in Y} P_{0c} = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$P(X = 1) = \sum_{c \in Y} P_{1c} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

3)**Conditional Probability**

Definition 7. *Conditional Probability is the probability of the classical state $b \in Y$ given that $X = a$. It is determined as-*

$$\Pr(Y = b \mid X = a) = \frac{\Pr((X, Y) = (a, b))}{\Pr(X = a)}$$

Example 11. *Using the data given in Example 10,*

$$\Pr(Y = 1 \mid X = 1) = \frac{\Pr(X \cap Y)}{\Pr(X)} = \frac{\Pr((X, Y) = (1, 1))}{\Pr(X = 1)} = \frac{1}{2}$$

$$\Pr(Y = 1 \mid X = 0) = \frac{\Pr(X \cap Y = (0, 1))}{\Pr(X = 0)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

Remark 6.

$$|\pi\rangle = \sum_{b \in Y} \left(\sum_{a \in X} p_{ab} \, |a\rangle\right) \otimes |b\rangle = \sum_{a \in X} |a\rangle \otimes \left(\sum_{b \in Y} p_{ab} \, |b\rangle\right)$$

4)**Case Example**

Given:

$$P(0,0) = \frac{1}{2}, \quad P(0,1) = \frac{1}{4}, \quad P(1,0) = \frac{1}{8}, \quad P(1,1) = \frac{1}{8}$$

For X

$$P(X=0) = P(0,0) + P(0,1) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$P(X=1) = P(1,0) + P(1,1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

For Y

$$P(Y=0) = P(0,0) + P(1,0) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}$$

$$P(Y=1) = P(0,1) + P(1,1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

Fixing $X = 0$

$$P(Y=1 \mid X=0) = \frac{P(0,1)}{P(X=0)} = \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}$$

$$P(Y=0 \mid X=0) = \frac{P(0,0)}{P(X=0)} = \frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$$

Fixing $Y = 1$

$$P(X=1 \mid Y=1) = \frac{P(1,1)}{P(Y=1)} = \frac{\frac{1}{8}}{\frac{3}{8}} = \frac{1}{3}$$

$$P(X=0 \mid Y=1) = \frac{P(0,1)}{P(Y=1)} = \frac{\frac{1}{4}}{\frac{3}{8}} = \frac{2}{3}$$

Check for Independence

For independence, we require:

$$P(a,b) = P(X=a) \cdot P(Y=b)$$

Check for $(0,0)$:

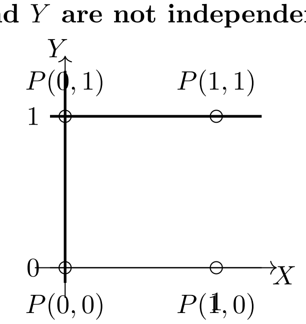
$$P(0,0) = \frac{1}{2}, \quad P(X=0)P(Y=0) = \frac{3}{4} \cdot \frac{5}{8} = \frac{15}{32}$$

Since:

$$\frac{1}{2} \neq \frac{15}{32}$$

Therefore, X and Y are not independent.

Fixing $Y = 1$



Fixing $(X = 0)$

Remark 7. *The independence of three systems is not transitive. For systems X, Y and Z - X and Y are independent and Y and Z are independent $\not\Rightarrow$ X and Z are independent.*

Example 12. *$X$is independent of Y, $Y$is independent of Z, but $X$is dependent of Z let*

$$P_X(0) = \frac{1}{2}, \quad P_X(1) = \frac{1}{2}; P_Y(0) = \frac{1}{2}, \quad P_Y(1) = \frac{1}{2}$$

And let

$$Z = X$$

Since $Z = X$ and Y are independent, we define:

$$P_{XYZ}(x,y,z) = \begin{cases} P_X(x)\, P_Y(y), & \text{if } z = x \\ 0, & \text{otherwise} \end{cases}$$

$$P(0,0,0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}; P(0,1,0) = \frac{1}{4}; P(1,0,1) = \frac{1}{4}; P(1,1,1) = \frac{1}{4}$$

All other probabilities are zero.

$$P_{XY}(x,y) = \sum_z P_{XYZ}(x,y,z) = P_X(x)P_Y(y)$$

Hence X and Y are independent.

$$P_{YZ}(y,z) = \sum_x P_{XYZ}(x,y,z) = P_Y(y)P_Z(z)$$

Hence Y and Z are independent.

$$P_{XZ}(0,0) = \frac{1}{2}$$

This is because when the value of x is fixed, it fixes the value of z. But

$$P_X(0)P_Z(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Since

$$P_{XZ}(0,0) \neq P_X(0)P_Z(0)$$

X and Z are not independent

2.3 Wednesday, March 25: Fifth lecture

1)**Quantum Multiple State Systems**

In a quantum system suppose,

$$\psi \in \mathscr{H}_1 \otimes \mathscr{H}_2 \otimes \cdots \otimes \mathscr{H}_n.$$

Then we can write

$$\psi = \sum_{i_1, i_2, \ldots, i_n} a_{i_1 i_2 \ldots i_n} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_n\rangle.$$

where the summation of the squares of the coefficients of each state is equal to 1, i.e.

$$\sum_{i_1, i_2, \ldots, i_n} |a_{i_1 i_2 \ldots i_n}|^2 = 1.$$

This corresponds to the idea that the norm of the quantum state vector, both for single and multiple systems, is equal to 1.

1)**Bell State**
A Bell state is a special entangled quantum state of two qubits. These states form a basis for the 2-qubit quantum space. There are four Bell states-
1. First Bell State

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle);\ \left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\1\end{bmatrix}$$

2. Second Bell State

$$\left|\Phi^-\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle);\ \left|\Phi^-\right\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\-1\end{bmatrix}$$

3. Third Bell State

$$\left|\Psi^+\right\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle);\ \left|\Psi^+\right\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\1\\0\end{bmatrix}$$

4. Fourth Bell State

$$\left|\Psi^-\right\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle);\ \left|\Psi^-\right\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}0\\1\\-1\\0\end{bmatrix}$$

Remark 8. *Many quantum states cannot be expressed as a tensor product of states of their individual subsystems. Such states are called non-separable, and this non-separability is precisely what defines Quantum Entanglement. Bell state is a primary example of this.*

Example 13. *Take two quantum subsystems with states $|\psi\rangle$ and $|\xi\rangle$ defined as:*

$$|\psi\rangle = \alpha_1|0\rangle + \beta_1|1\rangle, \tag{1}$$
$$|\xi\rangle = \alpha_2|0\rangle + \beta_2|1\rangle. \tag{2}$$

Suppose that the Bell state $|\Phi^+\rangle$ can be written in separable form:

$$|\Phi^+\rangle = |\psi\rangle \otimes |\xi\rangle. \tag{3}$$

The Bell state is given by:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right). \tag{4}$$

Expanding the tensor product:

$$\begin{aligned} |\psi\rangle \otimes |\xi\rangle &= (\alpha_1|0\rangle + \beta_1|1\rangle)(\alpha_2|0\rangle + \beta_2|1\rangle) && (5)\\ &= \alpha_1\alpha_2|00\rangle + \alpha_1\beta_2|01\rangle + \alpha_2\beta_1|10\rangle + \beta_1\beta_2|11\rangle. && (6)\end{aligned}$$

Equating coefficients with $|\Phi^+\rangle$, we obtain:

$$\alpha_1\alpha_2 = \frac{1}{\sqrt{2}}, \tag{7}$$
$$\beta_1\beta_2 = \frac{1}{\sqrt{2}}, \tag{8}$$
$$\alpha_1\beta_2 = 0, \tag{9}$$
$$\alpha_2\beta_1 = 0. \tag{10}$$

From $\alpha_1\beta_2 = 0$, either $\alpha_1 = 0$ or $\beta_2 = 0$. From $\alpha_2\beta_1 = 0$, either $\alpha_2 = 0$ or $\beta_1 = 0$. However, any such choice contradicts the conditions $\alpha_1\alpha_2 = \frac{1}{\sqrt{2}}$ and $\beta_1\beta_2 = \frac{1}{\sqrt{2}}$. Hence, no such factorization exists.

$$|\Phi^+\rangle \neq |\psi\rangle \otimes |\xi\rangle, \tag{11}$$

*so $|\Phi^+\rangle$ is a **non-separable (entangled) state**.*

Similarly, all Bell states are non-separable.

Remark 9. *Bell states are:*

Entangled- (non-separable), and

Linearly independent, forming an orthonormal basis of the two-qubit Hilbert space

2)**GHZ State**

$$\text{State: } \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$

3)**W state**

$$\text{State: } \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$$

Point to remember: The classical state system is divided into two types of vectors: independent vectors and dependent vectors. The quantum state system is divided into two types of vectors: entangled vectors and separable vectors.

EPR Paradox

Einstein, Podolsky, and Rosen introduced the EPR paradox as a challenge to the perceived completeness of quantum theory. Their thought experiment involves two particles that, after interacting, become separated yet remain linked by quantum entanglement. Quantum mechanics posits that a measurement of one particle instantly determines the state of the other, no matter how far apart they are. Einstein argued this conflicts with the principle of locality. Later developments, like Bell's Theorem, confirmed that such non-local correlations are real, supporting quantum mechanics.

2.4 Wednesday, April 8: Sixth lecture

Consider two quantum subsystems with state vectors $|\alpha\rangle$ and $|\beta\rangle$. The combined system is described by the tensor product state:

$$|\alpha\rangle \otimes |\beta\rangle .$$

Let M and N be linear operators acting on the first and second subsystems, respectively. The tensor product operator $M \otimes N$ acts on the composite system as follows:

$$(M \otimes N)(|\alpha\rangle \otimes |\beta\rangle) = (M |\alpha\rangle) \otimes (N |\beta\rangle).$$

This shows that the operator $M \otimes N$ acts independently on each subsystem: the operator M affects only the first state $|\alpha\rangle$, while N affects only the second state $|\beta\rangle$. Thus, the combined action is equivalent to applying each operator locally to its respective subsystem.

Proof

Consider two quantum systems with states $|\alpha\rangle \in \mathscr{H}_1$ and $|\beta\rangle \in \mathscr{H}_2$. The combined state is $|\alpha\rangle \otimes |\beta\rangle$.

Let M and N be linear operators acting on $\mathscr{H}_1$ and $\mathscr{H}_2$, respectively. We want to show:

$$(M \otimes N)(|\alpha\rangle \otimes |\beta\rangle) = (M |\alpha\rangle) \otimes (N |\beta\rangle).$$

Since quantum states live in vector spaces, we can expand them into their base state. Let $\{|i\rangle\}$ and $\{|j\rangle\}$ be bases for $\mathscr{H}_1$ and $\mathscr{H}_2$. Then:

$$|\alpha\rangle = \sum_i a_i |i\rangle , \qquad |\beta\rangle = \sum_j b_j |j\rangle .$$

Their tensor product would be as follows-

$$|\alpha\rangle \otimes |\beta\rangle = \left(\sum_i a_i |i\rangle \right) \otimes \left(\sum_j b_j |j\rangle \right) .$$

Now by using the property of tensor product that scalars factor out:

$$(cu) \otimes v = c(u \otimes v). \implies |\alpha\rangle \otimes |\beta\rangle = \sum_{i,j} a_i b_j (|i\rangle \otimes |j\rangle).$$

Now when we apply the operation $M \otimes N$ on the multiple system-

$$(M \otimes N)(|\alpha\rangle \otimes |\beta\rangle) = (M \otimes N) \left(\sum_{i,j} a_i b_j (|i\rangle \otimes |j\rangle) \right).$$

Since $M \otimes N$ is linear(it is a matrix that can be represented in the Kronecker form written as)-

$$M \otimes N = \begin{bmatrix} m_{11}N & m_{12}N & \cdots & m_{1n}N \\ m_{21}N & m_{22}N & \cdots & m_{2n}N \\ \vdots & \vdots & \ddots & \vdots \\ m_{m1}N & m_{m2}N & \cdots & m_{mn}N \end{bmatrix}.$$

we can bring it inside the sum:

$$= \sum_{i,j} a_i b_j (M \otimes N)(|i\rangle \otimes |j\rangle).$$

The definition of the tensor product operator is-

$$(M \otimes N)(|i\rangle \otimes |j\rangle) = (M\,|i\rangle) \otimes (N\,|j\rangle).$$

Thus,

$$= \sum_{i,j} a_i b_j \,(M\,|i\rangle \otimes N\,|j\rangle).$$

Now both $M\,|i\rangle$ and $N\,|j\rangle$ will give a column matrix which is explicitly written below-

$$M\,|i\rangle = \sum_k m_{ki}\,|k\rangle\,, \qquad N\,|j\rangle = \sum_\ell n_{\ell j}\,|\ell\rangle\,.$$

Thus we can further write our equation as-

$$= \sum_{i,j} a_i b_j \left(\sum_k m_{ki}\,|k\rangle \otimes \sum_\ell n_{\ell j}\,|\ell\rangle \right).$$

Using the bi linearity property of tensor product, we get-

$$= \sum_{i,j} a_i b_j \sum_{k,\ell} m_{ki} n_{\ell j} (|k\rangle \otimes |\ell\rangle).$$

By rearranging the sums according to the indexes taken-

$$= \sum_{k,\ell} \left(\sum_i m_{ki} a_i \right) \left(\sum_j n_{\ell j} b_j \right) (|k\rangle \otimes |\ell\rangle).$$

Now, the $\sum_i m_{ki} a_i$can be written back as

$$\sum_i m_{ki} a_i = (M\,|\alpha\rangle)_k, \qquad \sum_j n_{\ell j} b_j = (N\,|\beta\rangle)_\ell.$$

Hence,

$$= \sum_{k,\ell} (M\,|\alpha\rangle)_k (N\,|\beta\rangle)_\ell (|k\rangle \otimes |\ell\rangle).$$

This gives the tensor product:

$$(M\,|\alpha\rangle) \otimes (N\,|\beta\rangle).$$

Hence proved.

Remark 10. *The definition of the tensor product is defined for simple basis vectors of the subsystem. The proof generalizes it for any quantum state defined in that particular subsystem i.e any* $|\alpha\rangle \in \mathscr{H}_1$ *and* $|\beta\rangle \in \mathscr{H}_2$

2.5 Thursday, April 9: Seventh lecture

1)**Conjugate Symmetry of Inner Products in Dirac Notation**

Theorem 2. *For any two kets $|\psi\rangle$ and $|\phi\rangle$ in a quantum system, the inner product satisfies*

$$\langle \psi \mid \phi \rangle = \overline{\langle \phi \mid \psi \rangle}.$$

2)**Orthonormal basis Vectors and Orthogonal Vectors**
Two vectors $|v\rangle$ and $|w\rangle$ are said to be orthogonal if:

$$\langle v \mid w \rangle = 0.$$

More generally, for a set of vectors $\{|a_i\rangle\}$,

$$\langle a_i \mid a_j \rangle = 0 \quad \text{for all } i \neq j.$$

Let $\{|e_1\rangle, |e_2\rangle, |e_3\rangle, \ldots, |e_n\rangle\}$ be a set of orthogonal basis vectors. Then,

$$\langle e_i \mid e_k \rangle = \begin{cases} 1 & \textbf{if } i = k, \\ 0 & \textbf{if } i \neq k. \end{cases}$$

The orthogonal set of unit one basis vectors is called orthonormal set of basic vectors.

Lemma 2. *The cardinality of any orthogonal set of vectors in $\mathbb{C}^n$ is less than or equal to n.*

Proof: Let $\{v_1, v_2, \ldots, v_k\}$ be an orthogonal set of nonzero vectors in $\mathbb{C}^n$. So for $i \neq j$, we have

$$\langle v_i, v_j \rangle = 0.$$

Show that the vectors are linearly independent is important for the proof.
Suppose

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0.$$

Now take the inner product of both sides with v_j:

$$\left\langle \sum_{i=1}^{k} c_i v_i,\, v_j \right\rangle = 0.$$

Using linearity, this becomes:

$$\sum_{i=1}^{k} c_i \langle v_i, v_j \rangle = 0.$$

But since the vectors are orthogonal, all terms vanish except the one with $i = j$:

$$c_j \langle v_j, v_j \rangle = 0.$$

Since $v_j \neq 0$, we know $\langle v_j, v_j \rangle > 0$, so:

$$c_j = 0.$$

This works for every j, so all coefficients must be zero. Hence, the vectors are linearly independent.
Now we use the fact that in $\mathbb{C}^n$, there cannot be more than n linearly independent vectors. Therefore,

$$k \leq n.$$

Hence Proved.

3)**Some identities defined in Dirac notation**

Definition 8. *For vectors $|a\rangle, |b\rangle, |c\rangle, |d\rangle$, the identity:*

$$(\langle a| \otimes \langle b|)(|c\rangle \otimes |d\rangle) = \langle a \mid c\rangle \langle b \mid d\rangle.$$

holds true. Equivalently,

$$\langle a \otimes b \mid c \otimes d\rangle = \langle a \mid c\rangle \langle b \mid d\rangle.$$

Proof:

The property of tensor products that was defined in subsection 2.4 will be used to complete the proof-

$$(M \otimes N)(|\alpha\rangle \otimes |\beta\rangle) = (M |\alpha\rangle) \otimes (N |\beta\rangle).$$

The bra vector can be considered an linear operator acting on the ket vector such that it transforms the vector from the Hilbert Space to complex value in the Complex number system. Hence for the proof, we would take-

$$\langle a| : \mathscr{H} \to \mathbb{C}, \quad |x\rangle \mapsto \langle a|x\rangle.$$

Thus, we may treat $\langle a|$ and $\langle b|$ as operators.

$$M = \langle a|, \quad N = \langle b|, \quad |\alpha\rangle = |c\rangle, \quad |\beta\rangle = |d\rangle.$$

Then application of the tensor product property will give:

$$(\langle a| \otimes \langle b|)(|c\rangle \otimes |d\rangle) = (\langle a| |c\rangle) \otimes (\langle b| |d\rangle).$$

Since $\langle a| |c\rangle$ and $\langle b| |d\rangle$ are scalars in $\mathbb{C}$, a tensor product of the two scalar value will be just the multiplication of the two values.

$$\mathbb{C} \otimes \mathbb{C} \cong \mathbb{C}, \quad \lambda \otimes \mu = \lambda\mu,$$

Hence the conclusion stated is-

$$(\langle a| \otimes \langle b|)(|c\rangle \otimes |d\rangle) = \langle a \mid c\rangle \langle b \mid d\rangle.$$

Hence Proved.

Remark 11. *If A and C are two unitary operators then the tensor product of A and C is also unitary.*

Unitary Operators Preserve Inner Products and Norms(in a multiple subsystems criteria)

We use the identity:

$$(\langle a| \otimes \langle b|)(|c\rangle \otimes |d\rangle) = \langle a \mid c\rangle \langle b \mid d\rangle.$$

Let U_A and U_B be unitary operators,

$$U_A^\dagger U_A = I, \quad U_B^\dagger U_B = I.$$

Let

$$|\psi\rangle = |v\rangle \otimes |w\rangle, \quad |\phi\rangle = |x\rangle \otimes |y\rangle.$$

Apply the operator $U_A \otimes U_B$(using the tensor product rule):

$$|\psi'\rangle = (U_A \otimes U_B)|\psi\rangle = (U_A|v\rangle) \otimes (U_B|w\rangle),$$
$$|\phi'\rangle = (U_A|x\rangle) \otimes (U_B|y\rangle).$$

Preservation of Inner Products

$$\begin{aligned}
\langle\psi'|\phi'\rangle &= (\langle v|U_A^\dagger \otimes \langle w|U_B^\dagger)\big(U_A|x\rangle \otimes U_B|y\rangle\big) \\
&= \langle v|U_A^\dagger U_A|x\rangle \cdot \langle w|U_B^\dagger U_B|y\rangle \\
&= \langle v|x\rangle \cdot \langle w|y\rangle \\
&= \langle v \otimes w \mid x \otimes y\rangle \\
&= \langle \psi \mid \phi\rangle.
\end{aligned}$$

Preservation of Norms(by-product of preservation of Inner Products)

By taking $|\psi\rangle = |v\rangle \otimes |w\rangle$:

$$\begin{aligned}
\langle\psi'|\psi'\rangle &= \langle\psi|\psi\rangle \\
&= \langle v|v\rangle \cdot \langle w|w\rangle.
\end{aligned}$$

Thus, the norm is also preserved. This proves the **Remark 11** that states $U_A \otimes U_B$ is unitary.

2.6 Sunday, April 26: Seventh lecture

2.6.1 Some Logic Gates and Operators

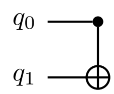
1)CNOT Operator in quantum systems
The Controlled-NOT (CNOT) gate is a fundamental two-qubit quantum gate. It consists of:

- A **control qubit**
- A **target qubit**

The rule of operation is:

- If the control qubit is $|0\rangle$, nothing happens.
- If the control qubit is $|1\rangle$, the target qubit is flipped.

The quantum circuit representation is as given below-



The $|qo\rangle$ and $|q1\rangle$ are taken now as $|0\rangle$ and $|0\rangle$ respectively. Firstly, we would change the control qubit with application of the Hadamard Gate in order to create superposition. Then CNOT can act on both possibilities 0 and 1 simultaneously. Applying a Hadamard gate to the control qubit, we get:

$$|0\rangle \xrightarrow{H} \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

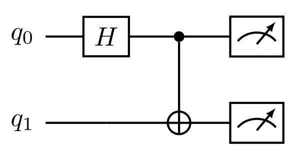
Entanglement using CNOT
Now apply CNOT to the system:

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes |0\rangle$$

After CNOT, we will get the output as-

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Given below is the circuit for better visualization-



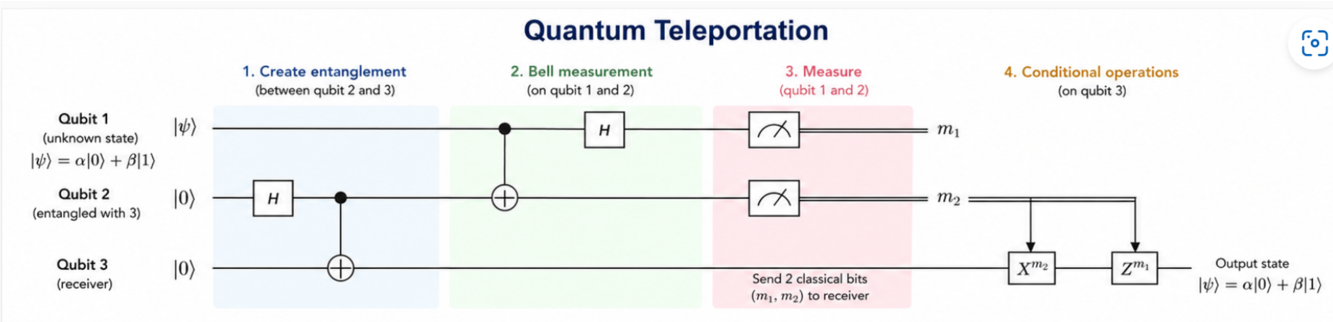
Now if we measure the resultant qubits, we would get-

- 50% probability of $|00\rangle$
- 50% probability of $|11\rangle$

States $|01\rangle$ and $|10\rangle$ will not occur. Thus CNOT gate creates entanglement, which acts as the quantum analogue of classical fan-out.

$$|00\rangle \xrightarrow{H\otimes I} \frac{|00\rangle + |10\rangle}{\sqrt{2}} \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

2.6.2 QUANTUM TELEPORTATION



So we would first explain the diagram, Initial state:

$$|q_2\rangle = |0\rangle, \quad |q_3\rangle = |0\rangle$$

Step 1. Apply Hadamard on Qubit 2.

$$|0\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Thus the combination of the system formed by qubit 2 and qubit 3 is-

$$\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

Now we shall apply CNOT with control as qubit 2 and target to be qubit 3. Therefore we will get-

$$|00\rangle \to |00\rangle, \quad |10\rangle \to |11\rangle$$

So the final state is:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

This is a Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Step 2: Bell Measurement

Now let our unknown state which we want to essentially teleport be-

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

First we would write this unknown state in tensor with the bell state(got by Step 1) and get the combined state-

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Expanding:

$$= \frac{1}{\sqrt{2}}\left(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle\right)$$

Now by application of CNOT keeping the control as Qubit 1 and target as Qubit 2), we are essentially modifing qubit 2 and qubit 3-

$$|000\rangle \to |000\rangle, \quad |011\rangle \to |011\rangle$$
$$|100\rangle \to |110\rangle, \quad |111\rangle \to |101\rangle$$

So the state becomes:

$$\frac{1}{\sqrt{2}}\left(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle\right)$$

Next, in this combined state we will apply Hadamard operator on qubit 1(which we took as control earlier) and write the final state- Using:

$$|0\rangle \to \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |1\rangle \to \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

After simplification:

$$= \frac{1}{2}\Big[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|0\rangle - \beta|1\rangle)$$
$$+|10\rangle(\beta|0\rangle + \alpha|1\rangle) + |11\rangle(\beta|0\rangle - \alpha|1\rangle)\Big]$$

Now we will measure the first two qubits (Qubit 1 and Qubit 2). We have destroyed the quantum superposition of these two qubits and have retrieved just the classical states from them. Now the possible outcomes that we would get for states of Qubit 3 are:

$$\begin{aligned}
|00\rangle &\to \alpha|0\rangle + \beta|1\rangle \\
|01\rangle &\to \alpha|0\rangle - \beta|1\rangle \\
|10\rangle &\to \beta|0\rangle + \alpha|1\rangle \\
|11\rangle &\to \beta|0\rangle - \alpha|1\rangle
\end{aligned}$$

After measurement, we obtain two classical bits:

$$(m_1, m_2) \in \{(0,0), (0,1), (1,0), (1,1)\}$$

These two measured bits are classical information that will be sent to the receiver(Person who holds the qubit 3). Depending on the measurement outcome, the following operations are done to Qubit 3:

$$\begin{aligned}
(0,0) &\to \alpha|0\rangle + \beta|1\rangle \\
(0,1) &\to \alpha|0\rangle - \beta|1\rangle \\
(1,0) &\to \beta|0\rangle + \alpha|1\rangle \\
(1,1) &\to \beta|0\rangle - \alpha|1\rangle
\end{aligned}$$

Where:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

After applying the appropriate correction by the receiver(person or computer), the final state of Qubit 3 will becomes:

$$\alpha|0\rangle + \beta|1\rangle$$

Thus, the original quantum state has been successfully teleported(now stored as qubit 3).

2.6.3 Deutsch Algorithm

Oracle Operator U_f

Definition 9. *Let $f : \{0,1\}^n \to \{0,1\}^m$ be a classical function. The corresponding quantum oracle U_f is defined as a unitary operator acting on computational basis states by:*

$$U_f\,|x\rangle|y\rangle = |x\rangle\,|y \oplus f(x)\rangle$$

where:

- $x \in \{0,1\}^n$ *is the input register,*
- $y \in \{0,1\}^m$ *is the output (target) register,*
- $\oplus$ *denotes bitwise addition modulo 2 (XOR).*

Remark 12. *The operator U_f extends linearly to arbitrary superpositions. For a general state:*

$$\sum_x \alpha_x |x\rangle|y\rangle$$

we have:

$$U_f\left(\sum_x \alpha_x |x\rangle|y\rangle\right) = \sum_x \alpha_x\, |x\rangle|y \oplus f(x)\rangle$$

In the Deutsch problem, we will consider functions from $\{0,1\}$ to $\{0,1\}$. There are exactly four such functions that can be constructed:

- $f_1(x) = 0$ (constant zero)
- $f_2(x) = 1$ (constant one)
- $f_3(x) = x$ (identity)
- $f_4(x) = 1 \oplus x$ (bit flip / NOT)

Example 14. *The computational basis that will be used in this example are ordered as:*

$$|00\rangle, |01\rangle, |10\rangle, |11\rangle$$

The matrix representation is showing oracle operations. Each column is representing one input and each row is representing one output.
1. $f(x) = 0$ (Constant Zero)

$$U_{f_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2. $f(x) = 1$ (Constant One)

$$U_{f_2} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

3. $f(x) = x$ (Identity)

$$U_{f_3} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

4. $f(x) = 1 \oplus x$ (NOT)

$$U_{f_4} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Main Working of Deutsch Algorithm

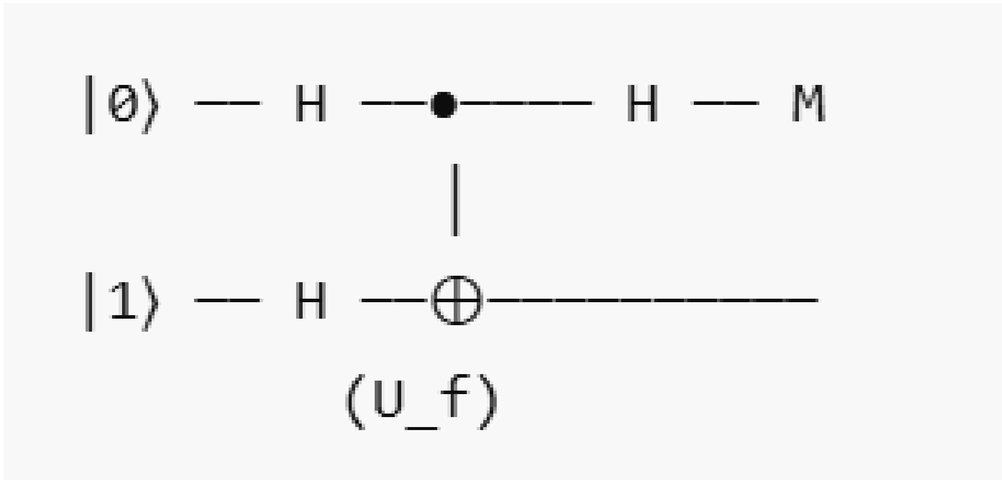


Figure 1: Circuit diagram for Deutsch Algorithm

The problem statement for deutsch algorithm is-
Let $f : \{0,1\} \to \{0,1\}$ be an unknown function(This can be out of the four functions that we described above). There are two types of such functions:

- **Constant:** $f(0) = f(1)$
- **Balanced:** $f(0) \neq f(1)$

The goal is to determine whether f is constant or balanced.
When using the **classical approach** evaluation of both:

$$f(0), \quad f(1)$$

is necessary to determine the type of function. However if we solve using quantum systems we just need to evaluate once and we will check further. The function is encoded into a unitary operator:

$$U_f|x, y\rangle = |x, y \oplus f(x)\rangle$$

The steps of the deutsch algorithm are as follows-

1. Initialize the state(use only the particular state):

$$|0\rangle \otimes |1\rangle$$

2. Apply Hadamard gates to both qubits:

$$\frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle - |1\rangle)$$

3. Apply the oracle U_f:

   Using phase kickback, the state becomes:

$$\frac{1}{2}\left[(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle\right](|0\rangle - |1\rangle)$$

Remark 13. *This remark explains phase kickback. Let $x \in \{0,1\}$ denote the value of the first qubit. We analyze the action of the oracle U_f on a basis state for a fixed x.]*
Case 1: $f(x) = 0$

$$U_f\left(|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}(|x, 0\rangle - |x, 1\rangle) = |x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

No change in the state.
Case 2: $f(x) = 1$

$$U_f\left(|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = \frac{1}{\sqrt{2}}\left(|x,1\rangle - |x,0\rangle\right) = -|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Thus, in general,

$$U_f\left(|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) = (-1)^{f(x)}\,|x\rangle\frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

This shows that the function value $f(x)$ is encoded as a phase factor $(-1)^{f(x)}$ on the first qubit, a phenomenon known as phase kickback.

4. Apply Hadamard to the first qubit:
   After applying the oracle, the first qubit is in the state:

$$\frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}}$$

   Applying the Hadamard transform:

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

   we get:

$$\frac{1}{2}\left[(-1)^{f(0)}(|0\rangle + |1\rangle) + (-1)^{f(1)}(|0\rangle - |1\rangle)\right] = \frac{1}{2}\left[\left((-1)^{f(0)} + (-1)^{f(1)}\right)|0\rangle + \left((-1)^{f(0)} - (-1)^{f(1)}\right)|1\rangle\right]$$

   Case 1: Constant function

   If $f(0) = f(1)$:

$$\Rightarrow (-1)^{f(0)} = (-1)^{f(1)}$$

$$\Rightarrow \text{Amplitude of } |1\rangle = 0 \quad \Rightarrow \quad \text{Output } |0\rangle$$

   Case 2: Balanced function
   If $f(0) \neq f(1)$:

$$\Rightarrow (-1)^{f(0)} = -(-1)^{f(1)}$$

$$\Rightarrow \text{Amplitude of } |0\rangle = 0 \quad \Rightarrow \quad \text{Output } |1\rangle$$

   The resulting state is:

$$\begin{cases} \pm|0\rangle & \text{if } f \text{ is constant} \\ \pm|1\rangle & \text{if } f \text{ is balanced} \end{cases}$$

5. Measure the first qubit:
   - Outcome 0 $\Rightarrow$ function is constant
   - Outcome 1 $\Rightarrow$ function is balanced

2.6.4 Quantum Advantage

Classical computation requires **2 evaluations** of f. Quantum computation requires **1 evaluation** of U_f. Thus, the Deutsch algorithm demonstrates a clear separation between classical and quantum computation.